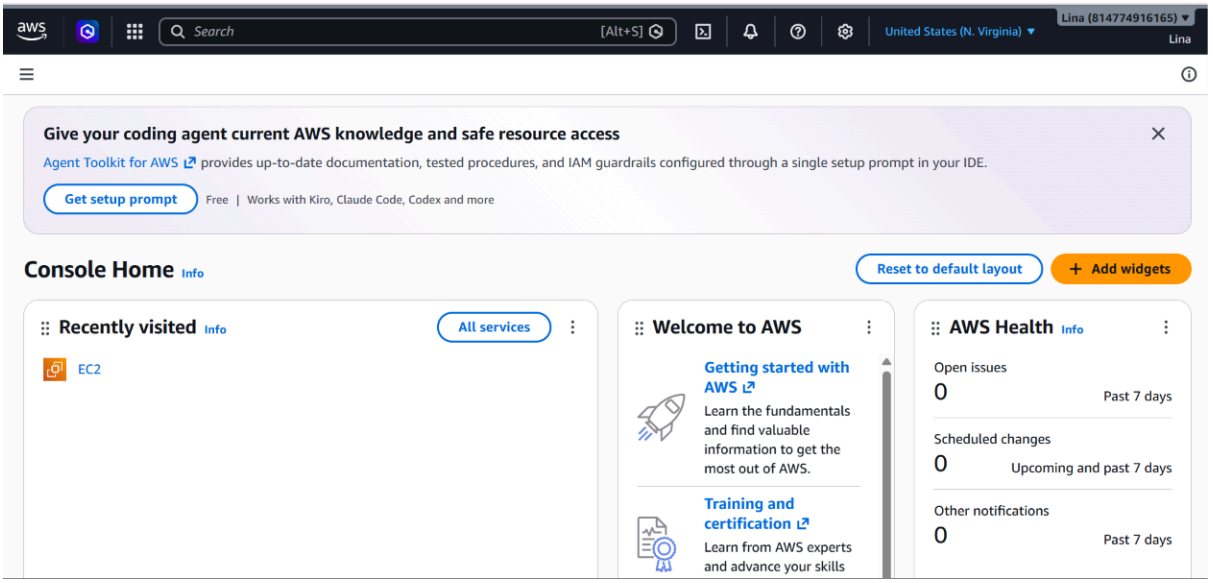
CLOUD COMPUTING TECHNOLOGIES LAB

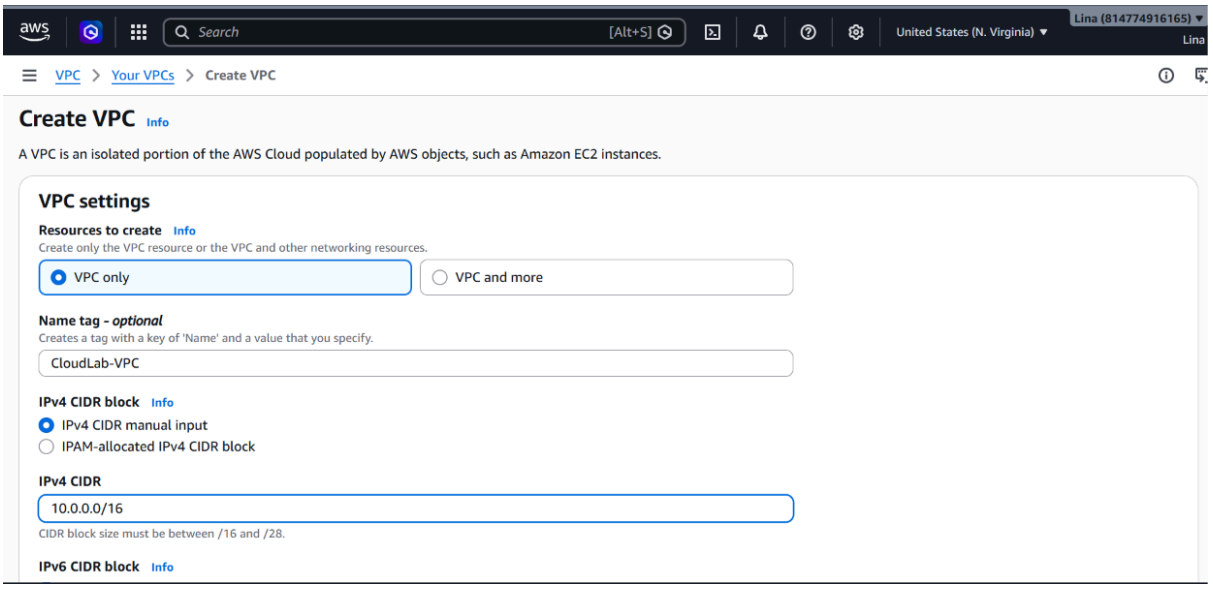
1.Select AWS Region



2.Create the VPC

AWS Console → VPC → Your VPCs → Create VPC

Name: CloudLab-VPC
IPv4 CIDR: 10.0.0.0/16



Click:

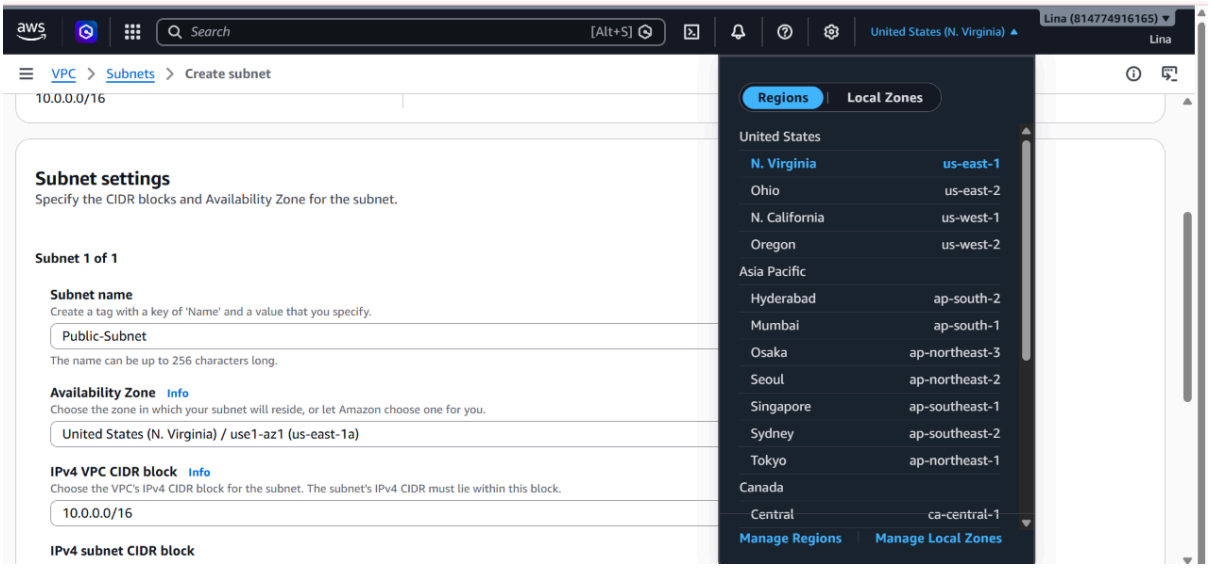
Create VPC

3. Create Public Subnet

Go to:

VPC → Subnets → Create subnet

VPC: CloudLab-VPC
Subnet name: Public-Subnet
Availability Zone: n.virginia
VPC CIDR: 10.0.1.0/24
Subnet CIDR: 10.0.1.0/24



4.Create Private Subnet

Go:

VPC → Subnets → Create subnet

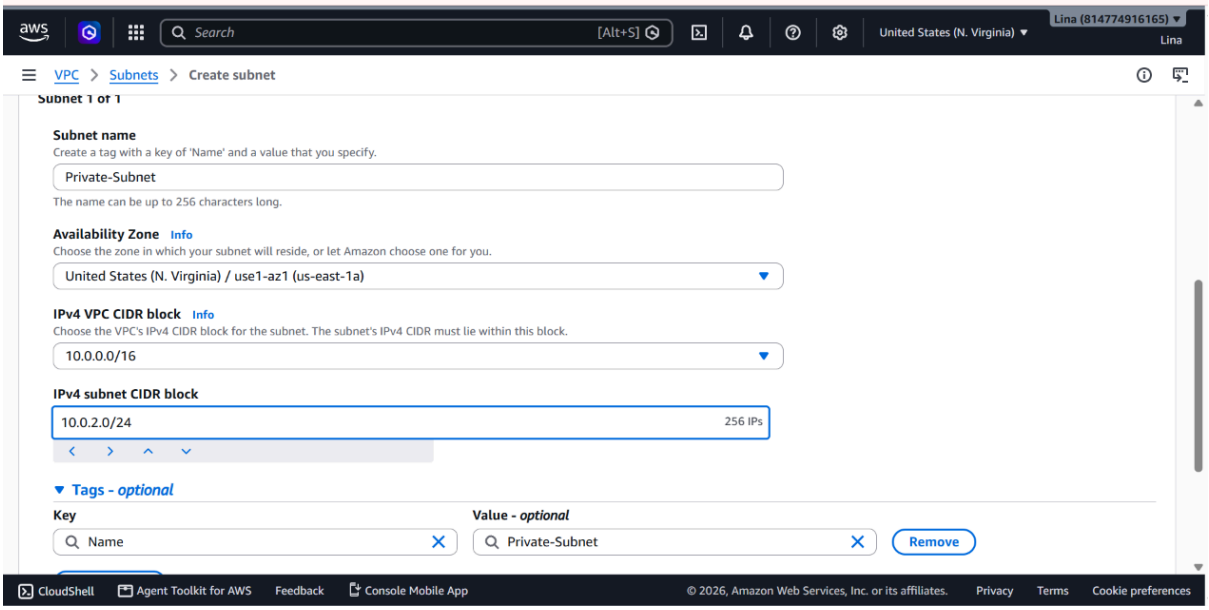
VPC:
CloudLab-VPC

Subnet name:
Private-Subnet

Availability Zone:
us-east-1a

IPv4 subnet CIDR:
10.0.2.0/24

Create subnet



5. Create Internet Gateway

VPC → Internet Gateways

Click:

Create internet gateway

Name:CloudLab-IGW

Go:

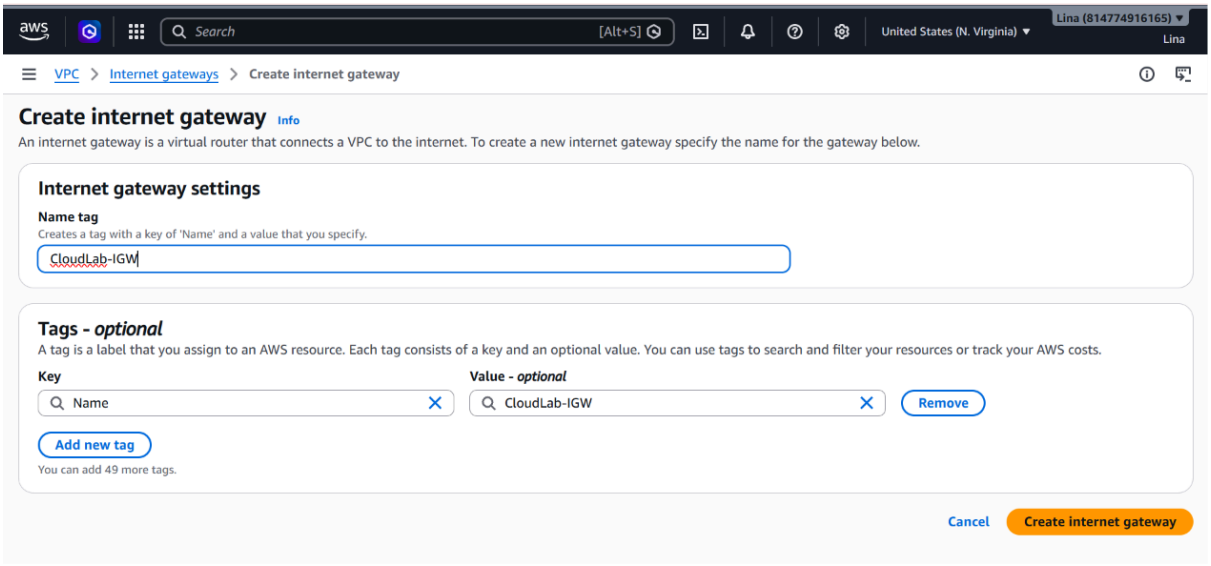
Actions → Attach to VPC

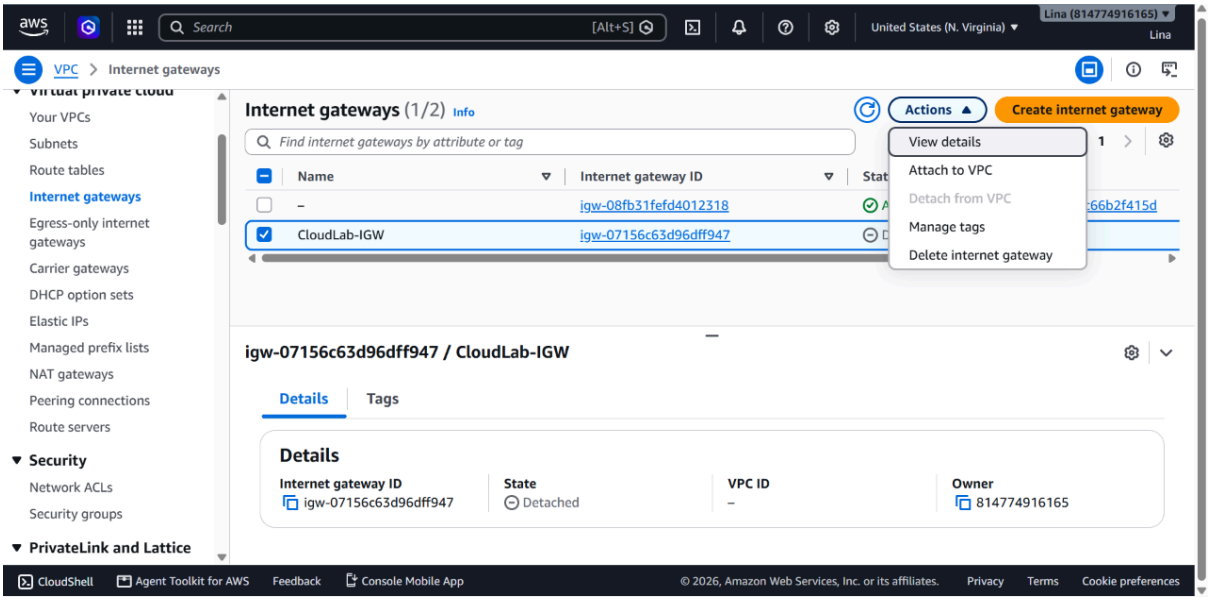
Choose:

CloudLab-VPC

Click:

Attach internet gateway





6.Create Public Route Table

Go:

VPC → Route Tables

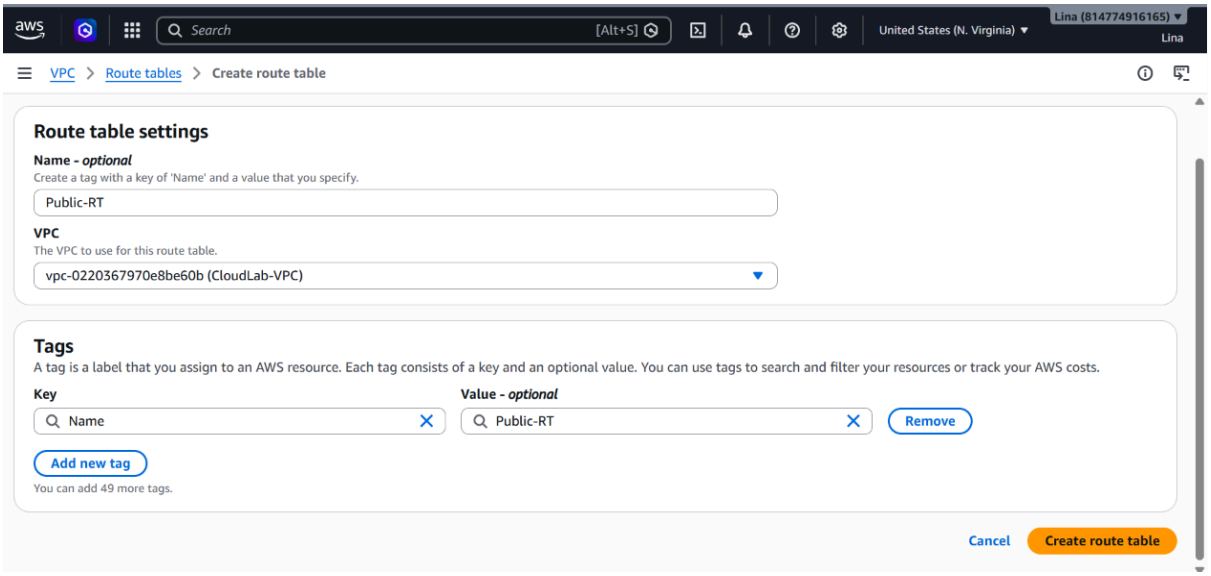
Click:

Create route table

Enter:Name:Public-RT

VPC:CloudLab-VPC

Click:**Create route table**



7. Add Internet Route

Select:

Public-RT

Go to:

Routes → Edit routes → Add route

Enter:

Destination:

0.0.0.0/0

Target:

Internet Gateway

Select:

CloudLab-IGW

Click:

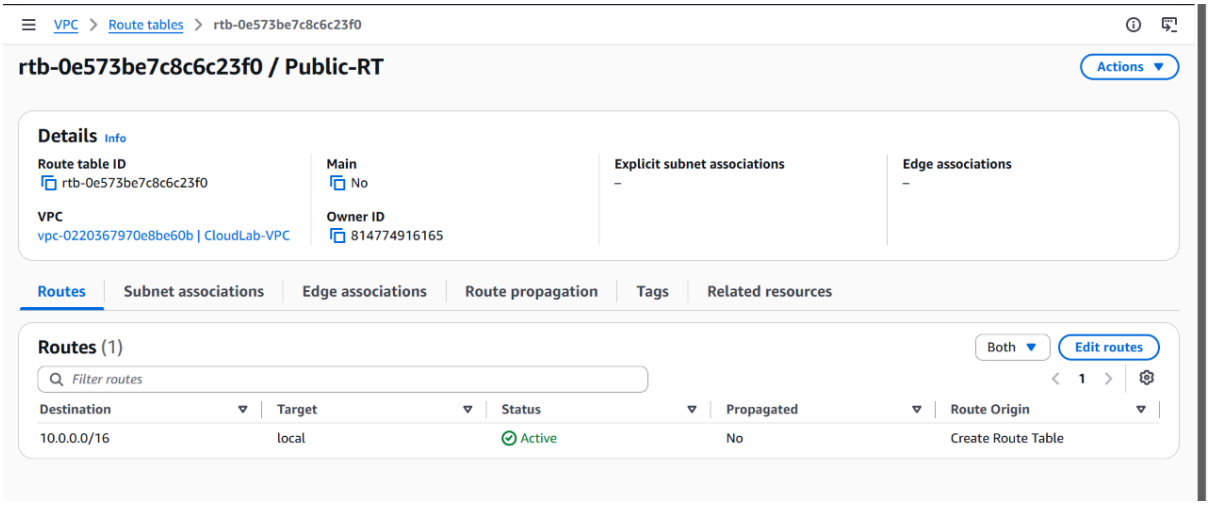
Save changes

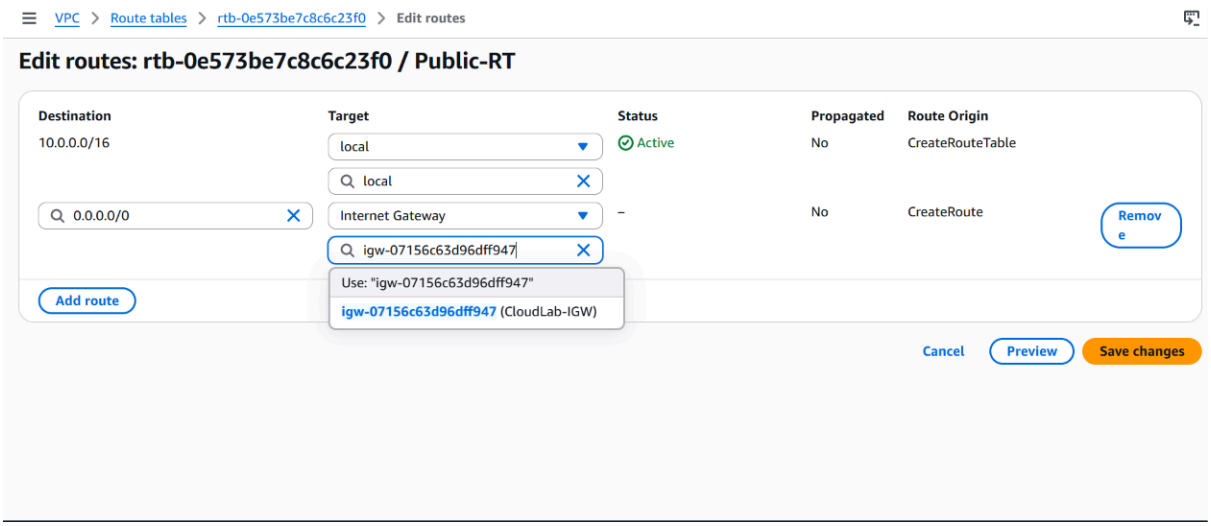
Your route is now:

0.0.0.0/0

↓

Internet Gateway





PART 8 — Associate Public Subnet

Still inside `Public-RT`.

Go:

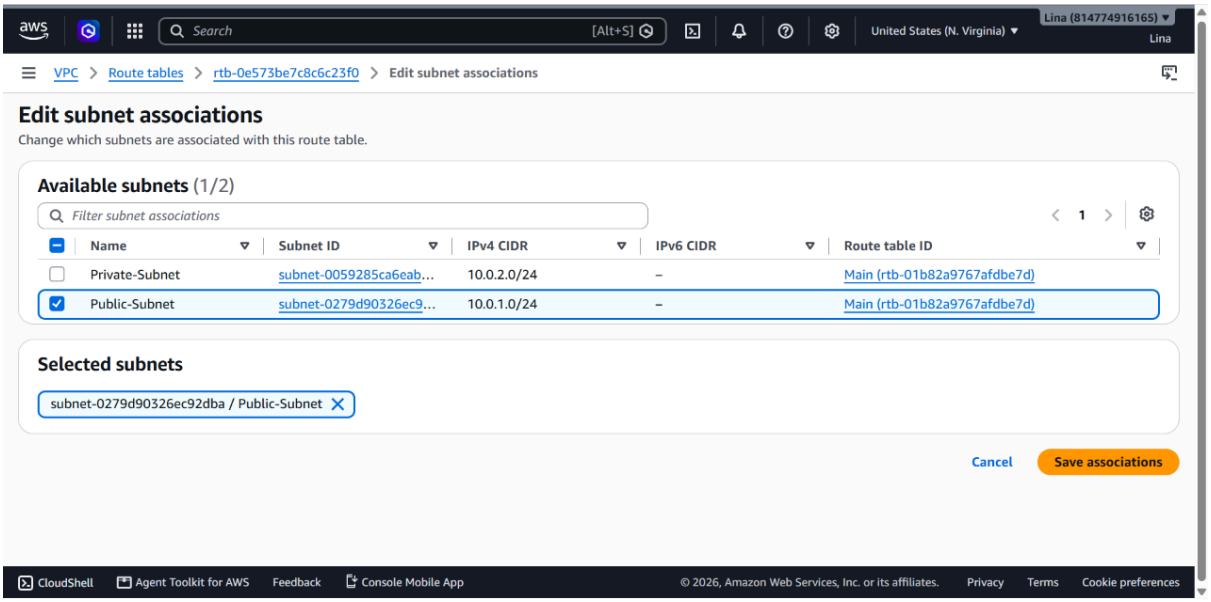
Subnet associations → Edit subnet associations

Select:

Public-Subnet

Click:

Save associations



PART 9 — Enable Public IPv4

Go:

VPC → Subnets → Public-Subnet

Click:

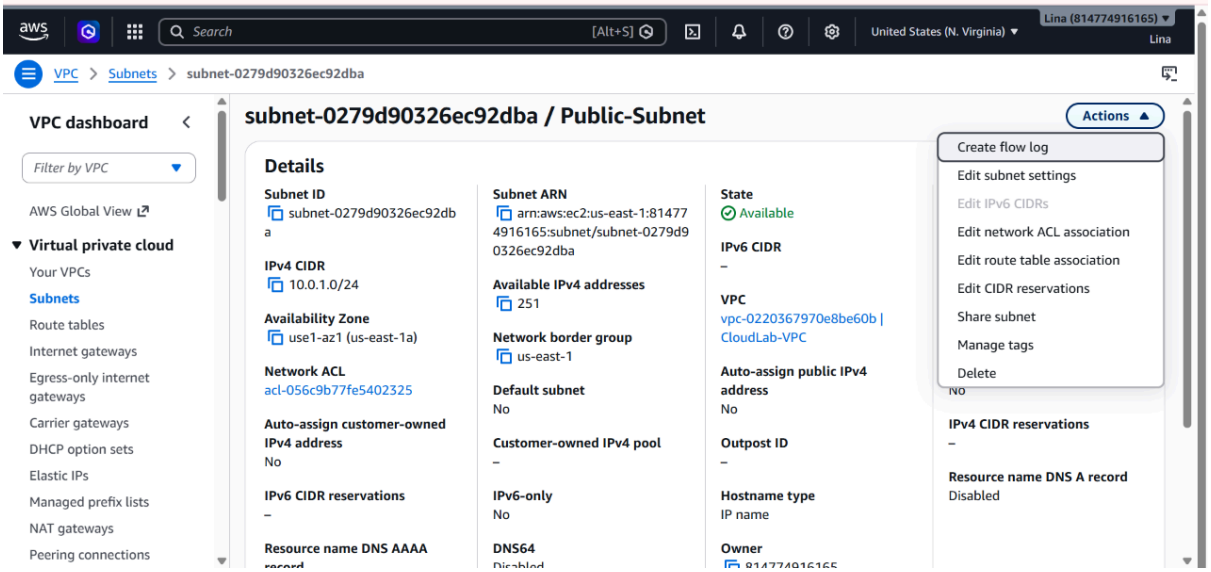
Actions → Edit subnet settings

Enable:

Enable auto-assign public IPv4 address

Click:

Save



PART 10 — Create S3 Bucket

Go:

AWS Console → S3

Click:

Create bucket

Use a unique name, for example:

cloudlab-reports-lina-2026-123456

Region

Make sure it says:

US East (N. Virginia)
us-east-1

Object Ownership

Leave default.

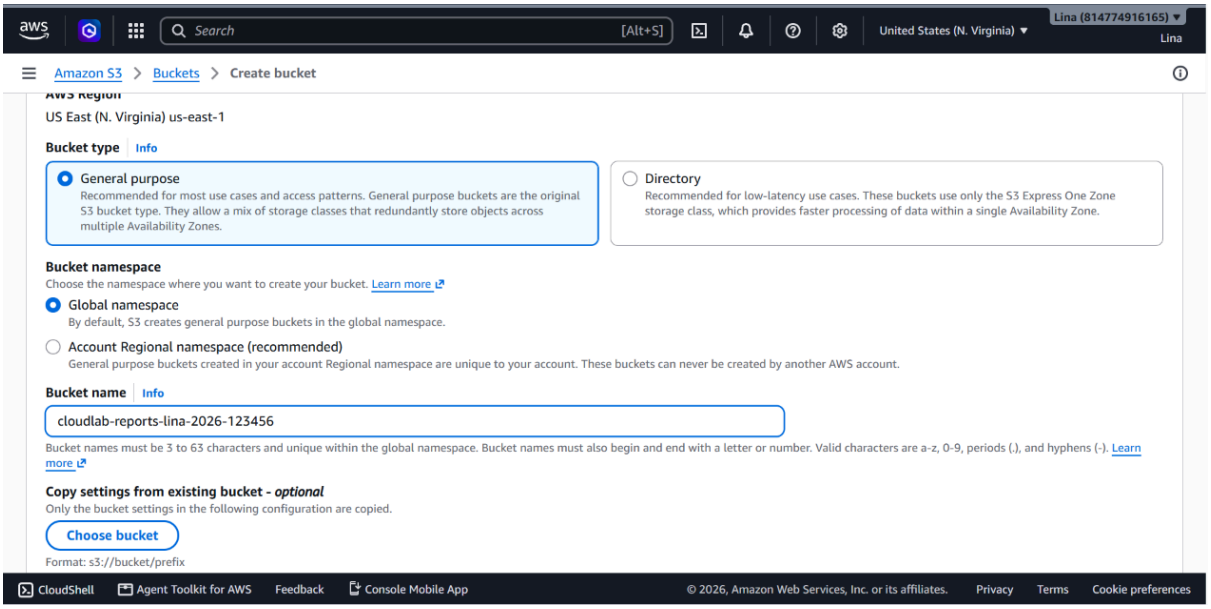
Block Public Access

Keep:

Block all public access
✓

Click:

Create bucket



11 — Create `reports` Folder

Open your bucket.

Click:

Create folder

Enter:

reports

Click:

Create folder

Open the folder.

Create a file on your computer:

sample.txt

Put:

AWS Cloud Infrastructure Lab

Upload it into:

reports/

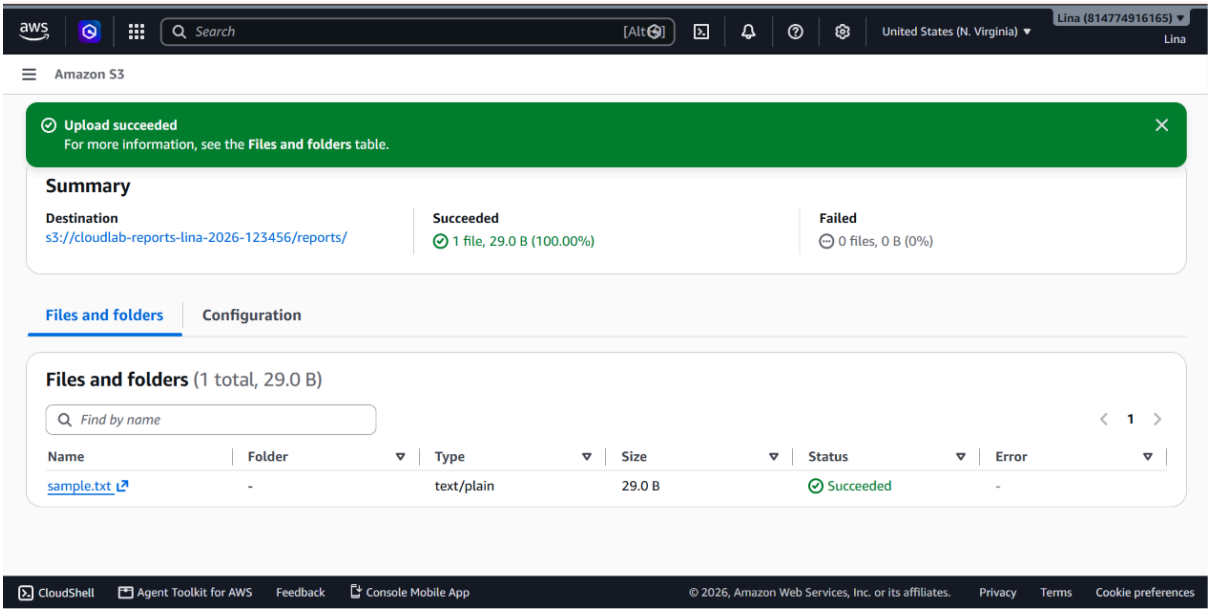
Your bucket:

cloudlab-reports-...

|

└── reports/

└── sample.txt



PART 12 — Create DynamoDB Table

Go:

AWS Console → DynamoDB

Click:

Create table

Enter:

Table name:

ReportsTable

Partition key:

ReportId

Type:

String

Select:

Capacity mode:

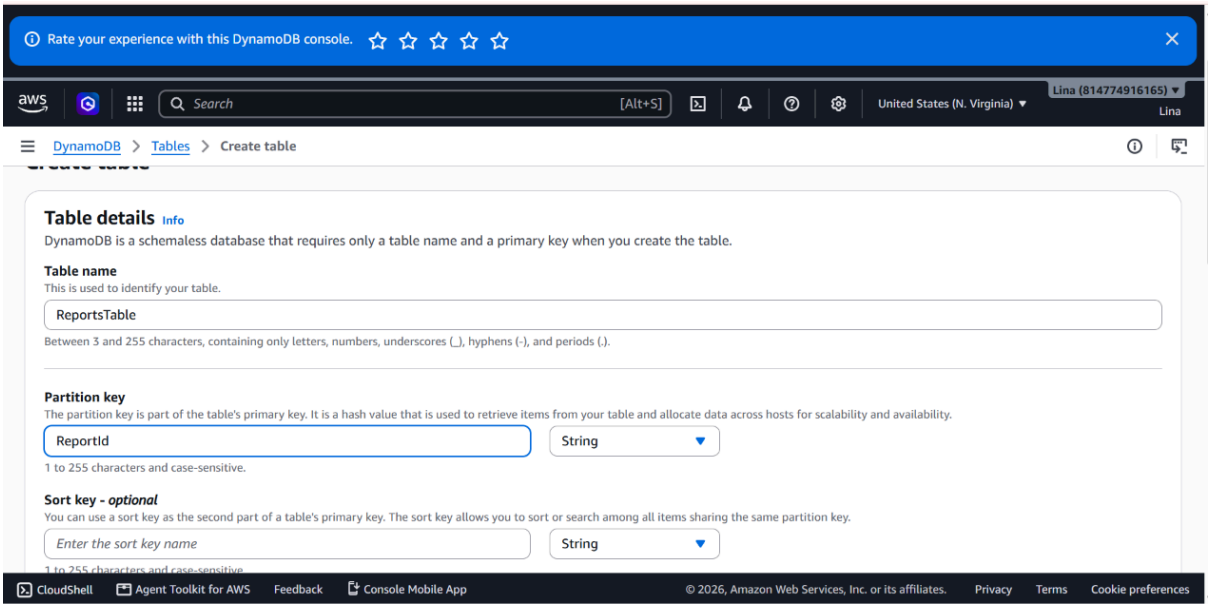
On-demand

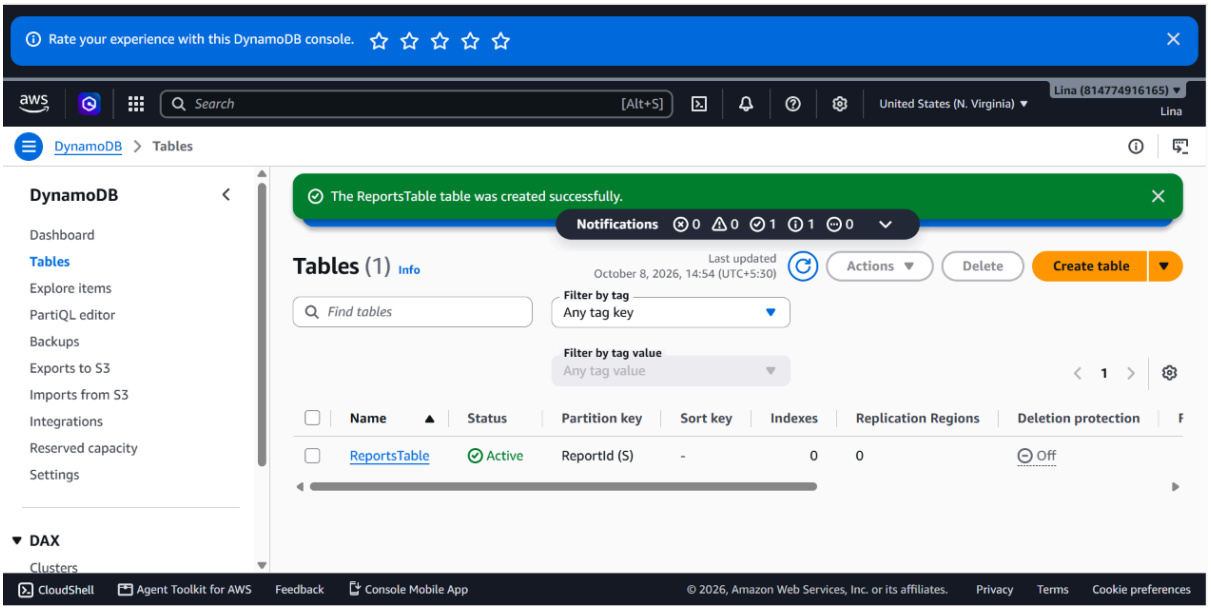
Click:

Create table

Wait until:

Status: Active





PART 13 — Create IAM Role

Go:

AWS Console → IAM → Roles

Click:

Create role

Trusted entity

Choose:

Trusted entity type:

AWS service

Use case:

EC2

Click:

Next

You don't need to attach AdministratorAccess.

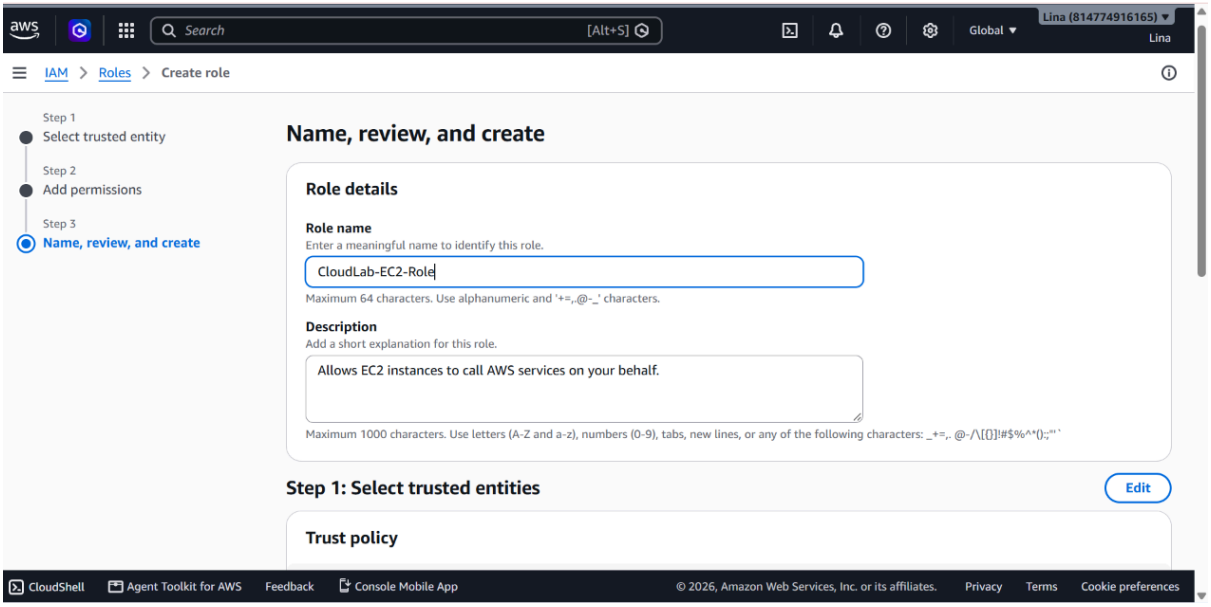
Create the role with:

Role name:

CloudLab-EC2-Role

Click:

Create role



PART 14 — Create Least-Privilege IAM Policy

Go:

IAM → Policies → Create policy

Select:

JSON

Paste:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadReportsOnly",
      "Effect": "Allow",
```

```
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/reports/*",
      "Condition": {
        "Bool": {
          "aws:SecureTransport": "true"
        }
      }
    },
    {
      "Sid": "WriteReportsTableOnly",
      "Effect": "Allow",
      "Action": [
        "dynamodb:PutItem"
      ],
      "Resource":
"arn:aws:dynamodb:us-east-1:YOUR-ACCOUNT-ID:table/ReportsTable"
    }
  ]
}
```

Replace:

YOUR-BUCKET-NAME

with your actual bucket name.

Example:

arn:aws:s3:::cloudlab-reports-lina-2026-123456/reports/*

Replace:

YOUR-ACCOUNT-ID

with your AWS account ID.

Your DynamoDB ARN becomes:

arn:aws:dynamodb:us-east-1:YOUR_ACCOUNT_ID:table/ReportsTable

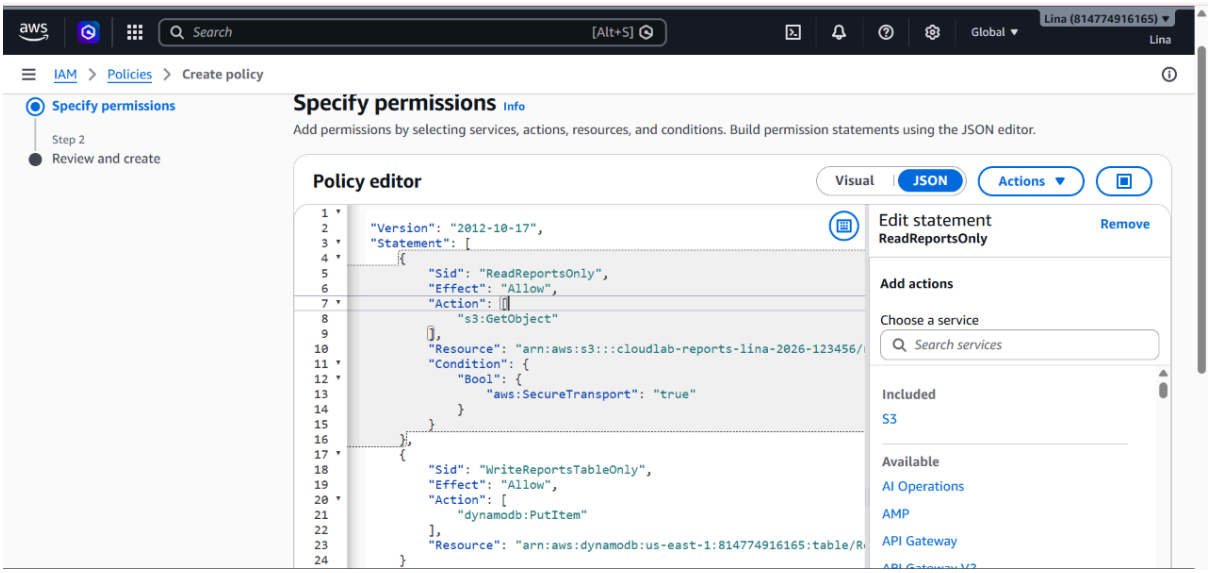
Click:

Next

Policy name:

CloudLab-LeastPrivilege-Policy

Create policy.



15 — Attach Policy to Role

Go:

IAM → Roles → CloudLab-EC2-Role

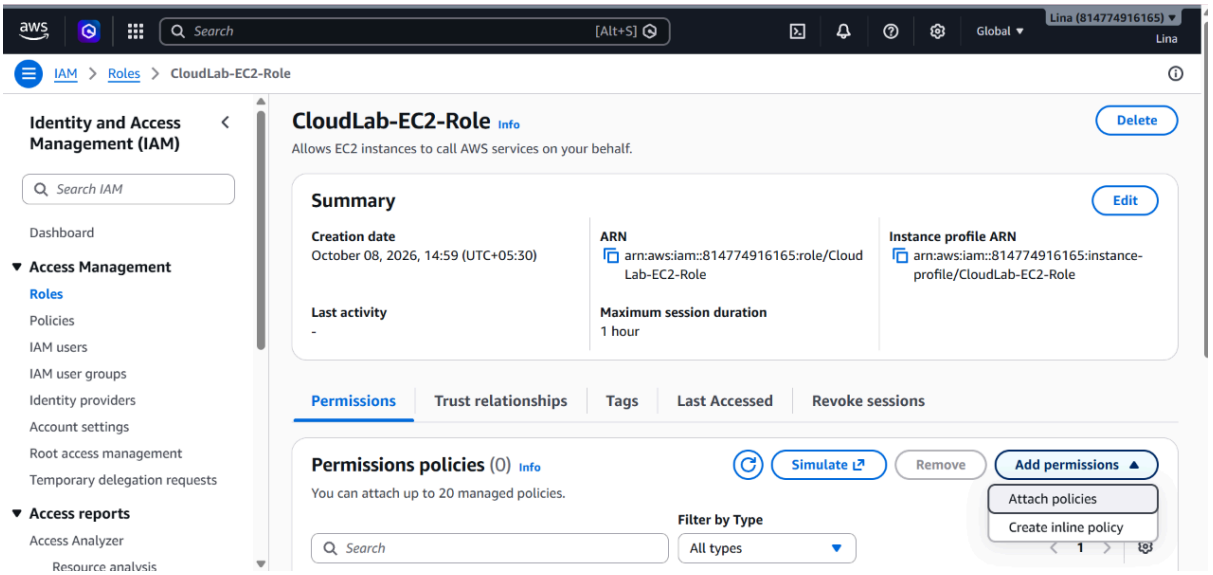
Click:

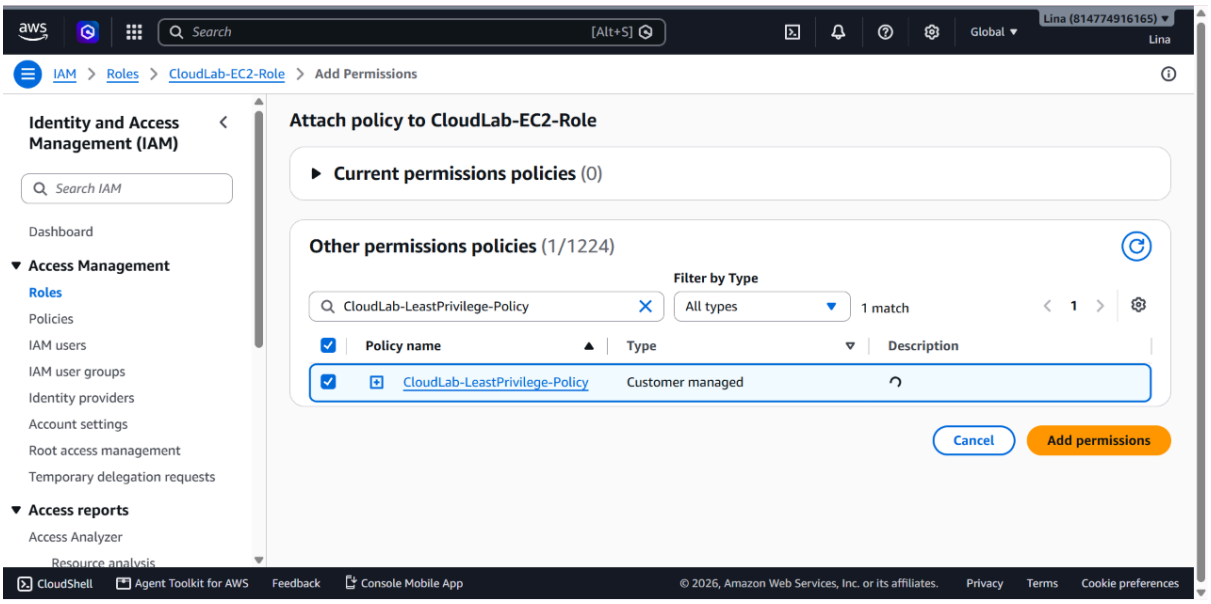
Add permissions → Attach policies

Select:

CloudLab-LeastPrivilege-Policy

Attach it.





16 — Create Security Group

Go:

EC2 → Security Groups

Click:

Create security group

Enter:

Security group name:

CloudLab-SG

Description:

Security group for CloudLab application

VPC:

CloudLab-VPC

Inbound rule

Add:

Type: SSH

Port: 22

Source: My IP

This is safer than:

0.0.0.0/0

Outbound

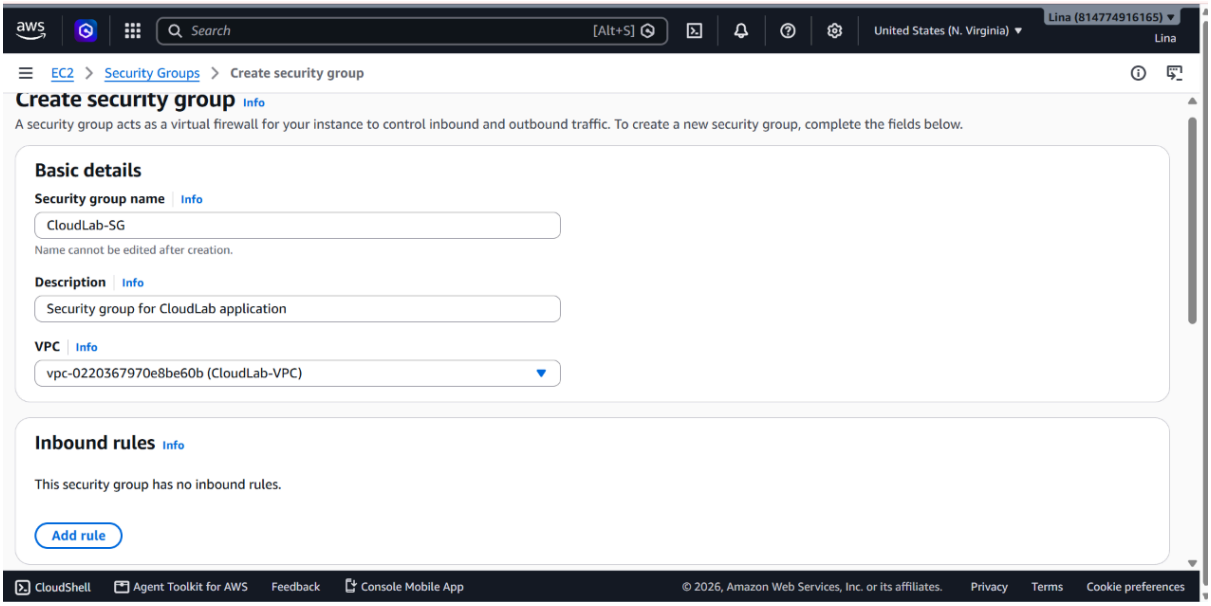
Keep:

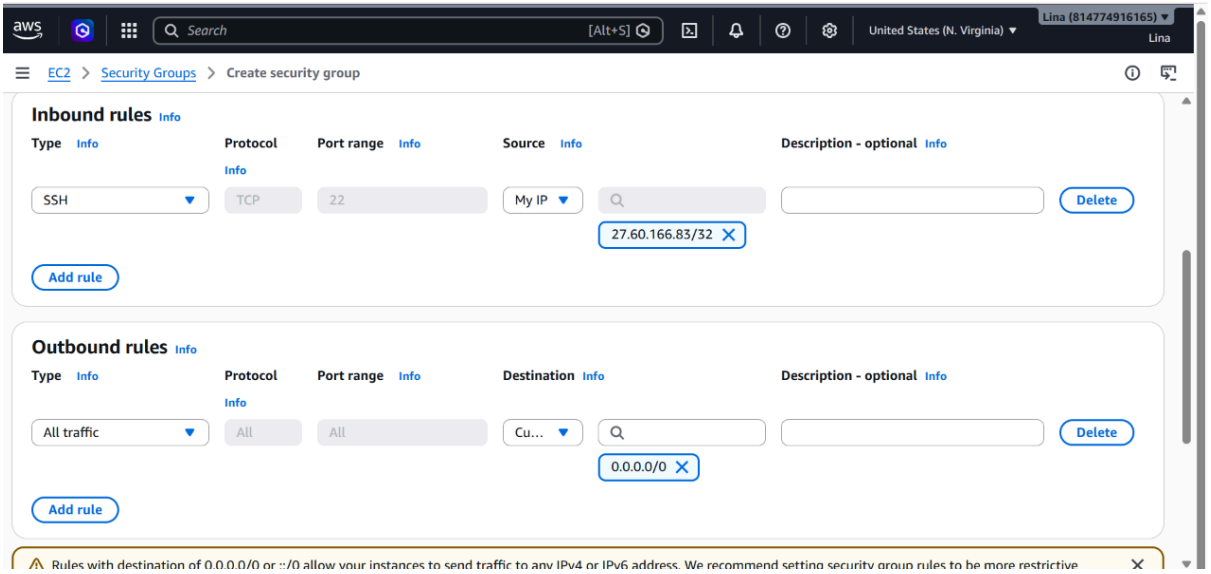
All traffic

0.0.0.0/0

Click:

Create security group





PART 17 — Launch EC2

Go:

EC2 → Instances → Launch instance

Name:

CloudLab-Application-Server

AMI

Choose:

Amazon Linux 2023

Instance type

Choose a small/free-tier eligible instance available in your account.

For example:

t2.micro

Key pair

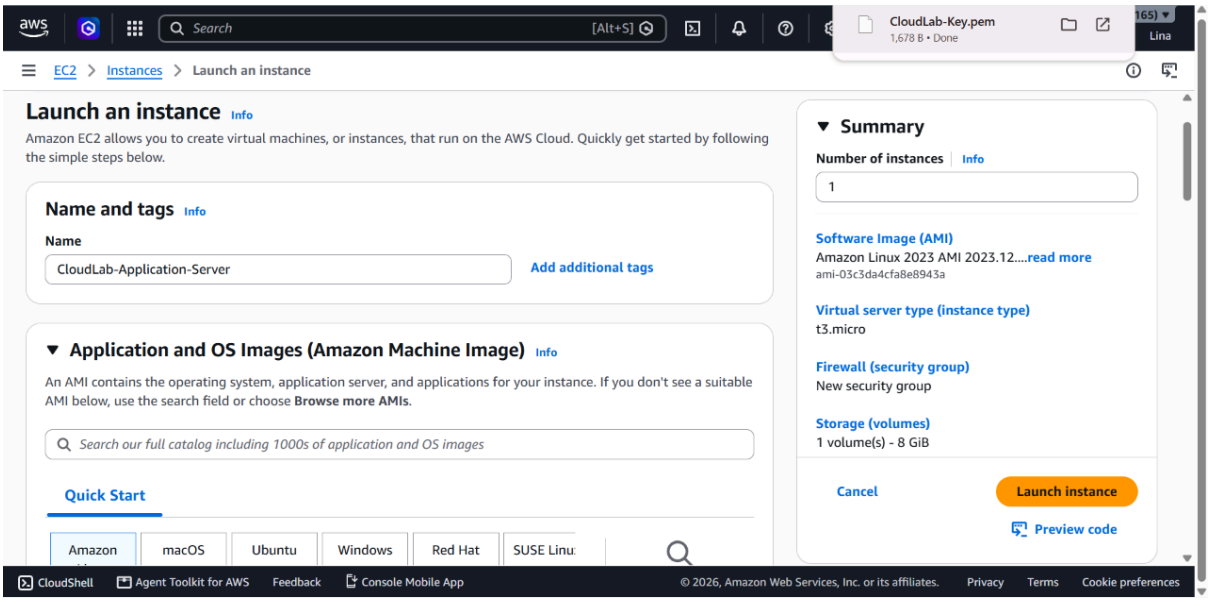
Create/select an EC2 key pair.

For example:

CloudLab-Key

Download the `.pem` file.

Keep this file safe. Never upload it to GitHub.



18 — Configure EC2 Network

Under Network settings:

VPC:

CloudLab-VPC

Subnet:

Public-Subnet

Auto-assign Public IP:

Enable

Security group:

CloudLab-SG